

WESTERN ISLANDS, ON, Canada

WMO: 714380

Lat: 45.036N

Lon: 80.358W

Elev: 191

StdP: 99.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.1	-16.1	-22.9	0.5	-17.9	-20.5	0.6	-15.4	21.3	-3.8	19.2	-2.8	7.6	0	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	24.7	20.4	23.6	19.9	22.6	19.4	21.8	23.4	21.0	22.7	20.2	22.0	5.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.3	22.8	20.4	15.4	22.1	19.6	14.6	21.6	64.6	23.5	61.8	22.8	59.1	22.0	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.5	15.0	DB	-22.3	27.7	2.8	1.3	-24.3	28.6	-25.9	29.4	-27.5	30.1	-29.5	31.0	
			WB	-22.6	22.9	2.6	1.0	-24.4	23.7	-25.9	24.3	-27.4	24.9	-29.2	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-5.0	-5.3	-2.1	3.5	9.6	15.3	18.6	19.2	16.3	10.7	4.1	-1.2
	DBStd	9.77	5.79	5.40	4.91	3.76	4.06	3.63	3.09	2.93	3.54	3.86	4.81	5.21
	HDD10.0	2100	466	428	376	198	58	4	0	0	2	38	183	348
	HDD18.3	4251	724	662	633	445	273	104	35	25	80	237	426	606
	CDD10.0	1014	0	0	1	3	45	161	265	284	189	60	7	0
	CDD18.3	126	0	0	0	0	1	12	42	51	18	1	0	0
	CDH23.3	123	0	0	0	0	4	17	44	44	14	0	0	0
	CDH26.7	4	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	7.8	9.8	8.7	7.8	7.4	6.9	6.0	6.1	6.3	6.9	8.8	9.5	9.6
Precipitation	PrecAvg	991	73	54	57	82	90	84	84	94	100	109	88	77
	PrecMax	1171	120	112	139	147	156	141	137	168	192	177	138	155
	PrecMin	811	38	20	8	31	33	29	38	47	38	48	41	33
	PrecStd	97	21	24	28	29	31	28	28	28	35	35	25	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	6.5	12.9	16.5	23.2	25.4	26.2	26.1	25.2	20.6	14.6	10.9
		MCWB	6.9	5.0	9.3	13.2	16.7	19.1	20.8	21.1	20.5	16.7	12.1	9.9
	2%	DB	5.4	4.5	8.5	13.0	20.5	23.2	24.6	24.5	23.3	18.7	12.7	8.4
		MCWB	4.3	3.2	6.1	9.1	14.4	18.3	20.6	20.9	19.9	15.7	10.5	7.2
	5%	DB	3.7	2.9	5.3	10.5	18.0	21.6	23.4	23.4	22.1	17.2	11.9	6.6
		MCWB	2.7	1.6	3.3	6.9	13.5	17.2	20.0	20.3	19.5	14.7	9.3	5.0
	10%	DB	2.3	1.6	3.2	8.4	15.7	20.2	22.3	22.5	20.9	15.8	11.4	5.4
		MCWB	1.4	0.6	1.6	5.3	11.9	17.0	19.3	19.7	18.5	13.6	9.2	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	5.7	10.3	14.7	18.4	21.2	22.9	23.0	21.9	17.7	13.0	10.3
		MCDB	8.0	6.0	12.5	16.1	22.0	23.3	24.3	24.5	23.8	19.4	13.9	11.1
	2%	WB	4.5	3.3	6.0	10.0	16.2	19.8	21.7	21.9	21.0	16.3	11.2	7.4
		MCDB	5.1	4.3	8.3	11.6	18.7	21.7	23.3	23.4	22.5	18.1	12.4	8.3
	5%	WB	2.7	1.9	3.5	7.6	14.2	18.9	20.8	21.1	20.1	15.1	9.6	5.3
		MCDB	3.6	2.6	4.9	9.7	17.0	20.7	22.5	22.8	21.8	16.6	11.5	6.4
	10%	WB	1.4	0.6	1.7	5.6	12.6	17.7	20.0	20.2	18.9	13.8	8.9	3.9
		MCDB	2.2	1.6	2.8	7.8	14.9	19.6	21.8	22.0	20.7	15.6	11.2	5.2
Mean Daily Temperature Range	5% DB	MDBR	6.1	5.8	5.5	5.6	6.9	6.0	3.7	3.2	3.9	4.1	4.2	4.8
		MCDBR	6.6	5.6	7.5	9.7	10.1	7.4	4.9	4.3	4.7	5.5	4.2	5.1
		MCWBR	6.3	5.2	5.7	6.7	6.5	4.4	3.3	3.4	3.8	4.4	4.0	4.9
	5% WB	MCDBR	6.7	5.9	7.6	9.4	9.4	6.4	4.6	4.1	4.8	5.1	5.3	4.8
		MCWBR	6.5	5.9	6.3	7.7	7.0	4.6	3.8	3.7	4.2	4.7	5.3	4.9
Clear-Sky Solar Irradiance	taub	0.287	0.293	0.312	0.350	0.377	0.402	0.406	0.402	0.367	0.348	0.315	0.286	
	taud	2.436	2.364	2.376	2.366	2.389	2.340	2.363	2.363	2.446	2.461	2.502	2.496	
	Ebn at Noon	834	895	923	911	892	867	859	849	853	821	795	801	
	Edh at Noon	71	91	104	115	116	123	119	114	97	83	67	61	
All-Sky Solar Radiation	RadAvg	1.21	2.13	3.52	4.58	5.46	5.98	6.06	5.15	3.98	2.32	1.29	0.86	
	RadStd	0.15	0.24	0.36	0.41	0.52	0.43	0.32	0.28	0.38	0.26	0.20	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page